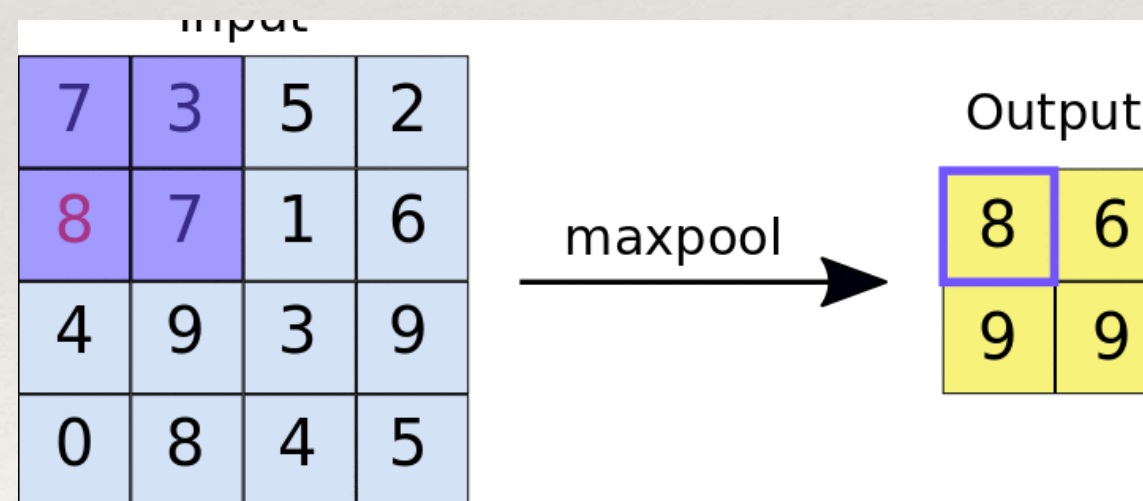
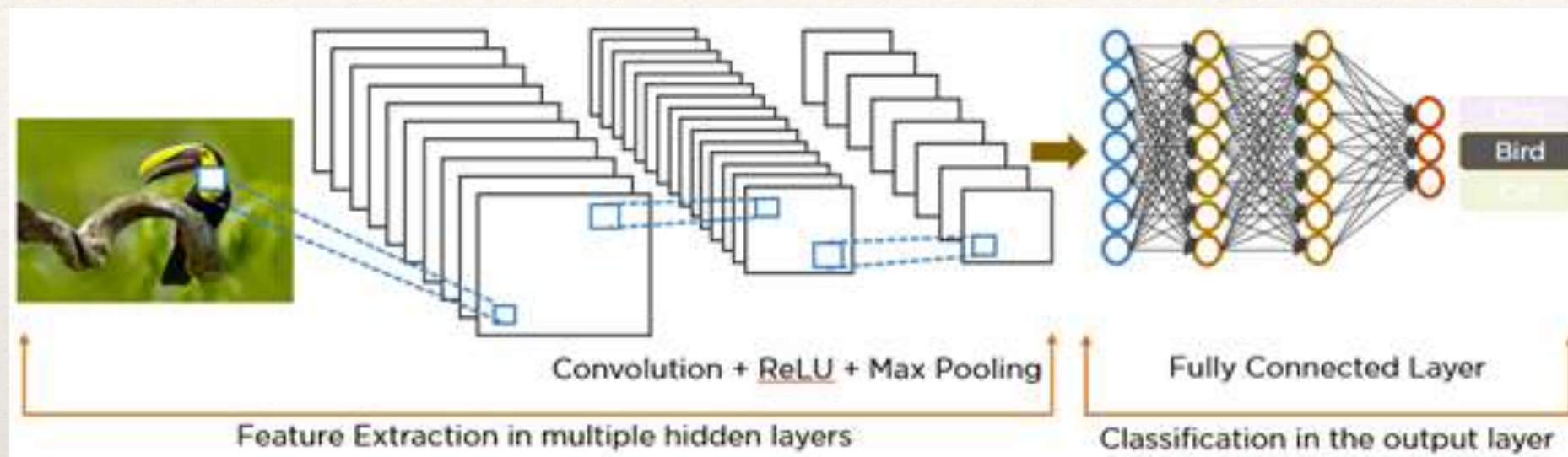


Deep Learning: Theory and Practice

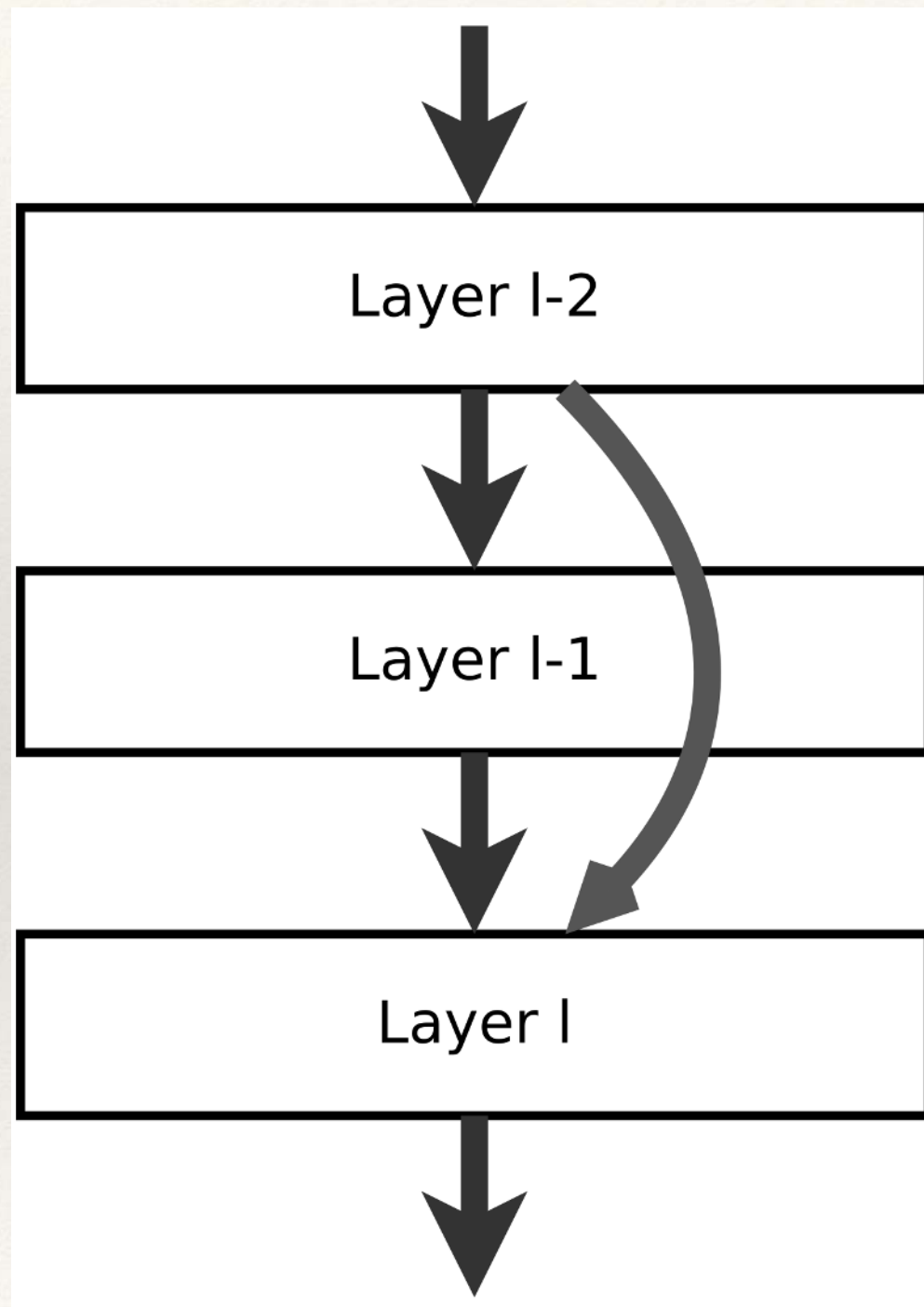
Advanced Topics in Deep Learning

11-04-2019

Back propagation in Pooling Layers

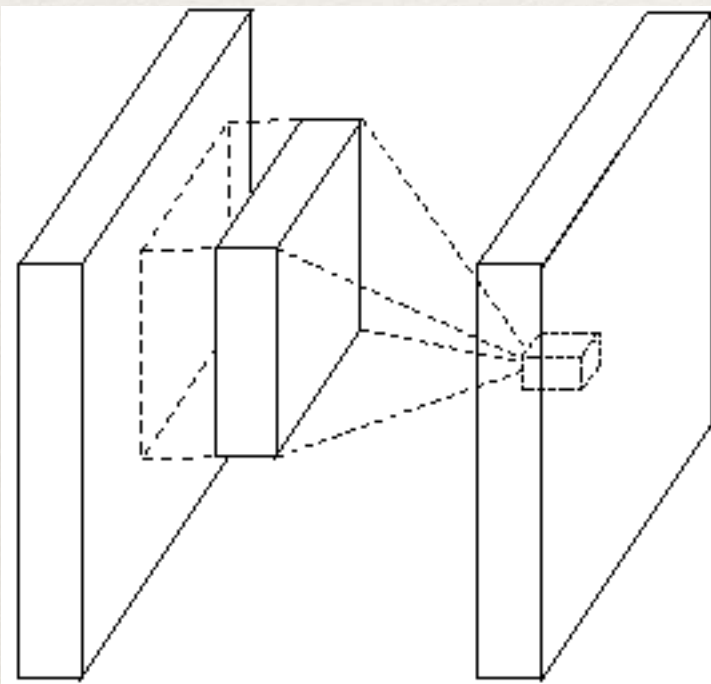


Resnet Architecture

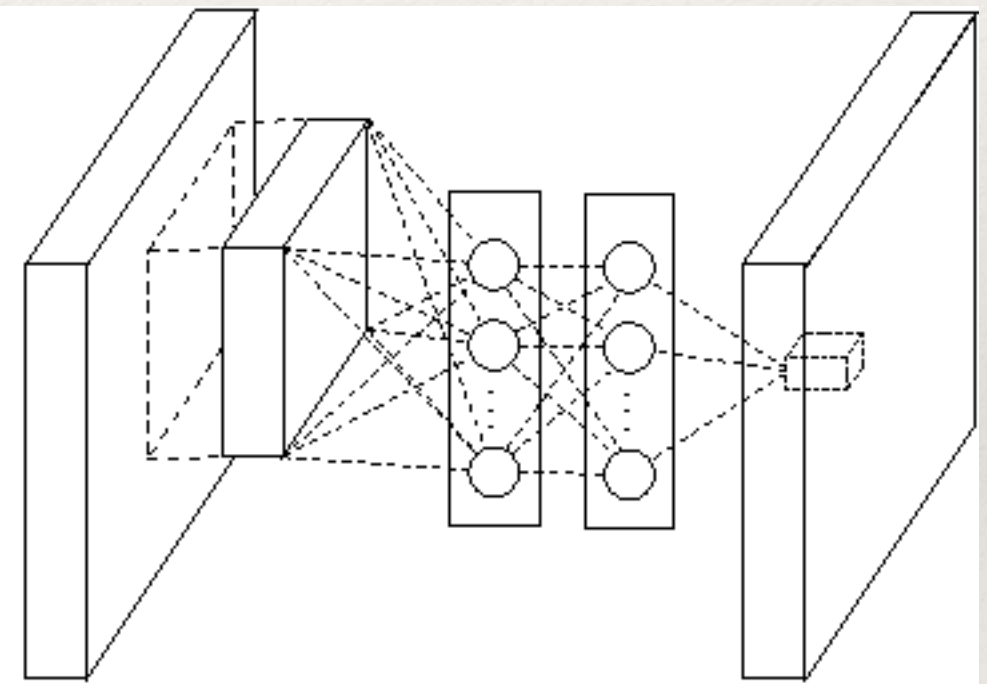


With skip connections

Network in Network

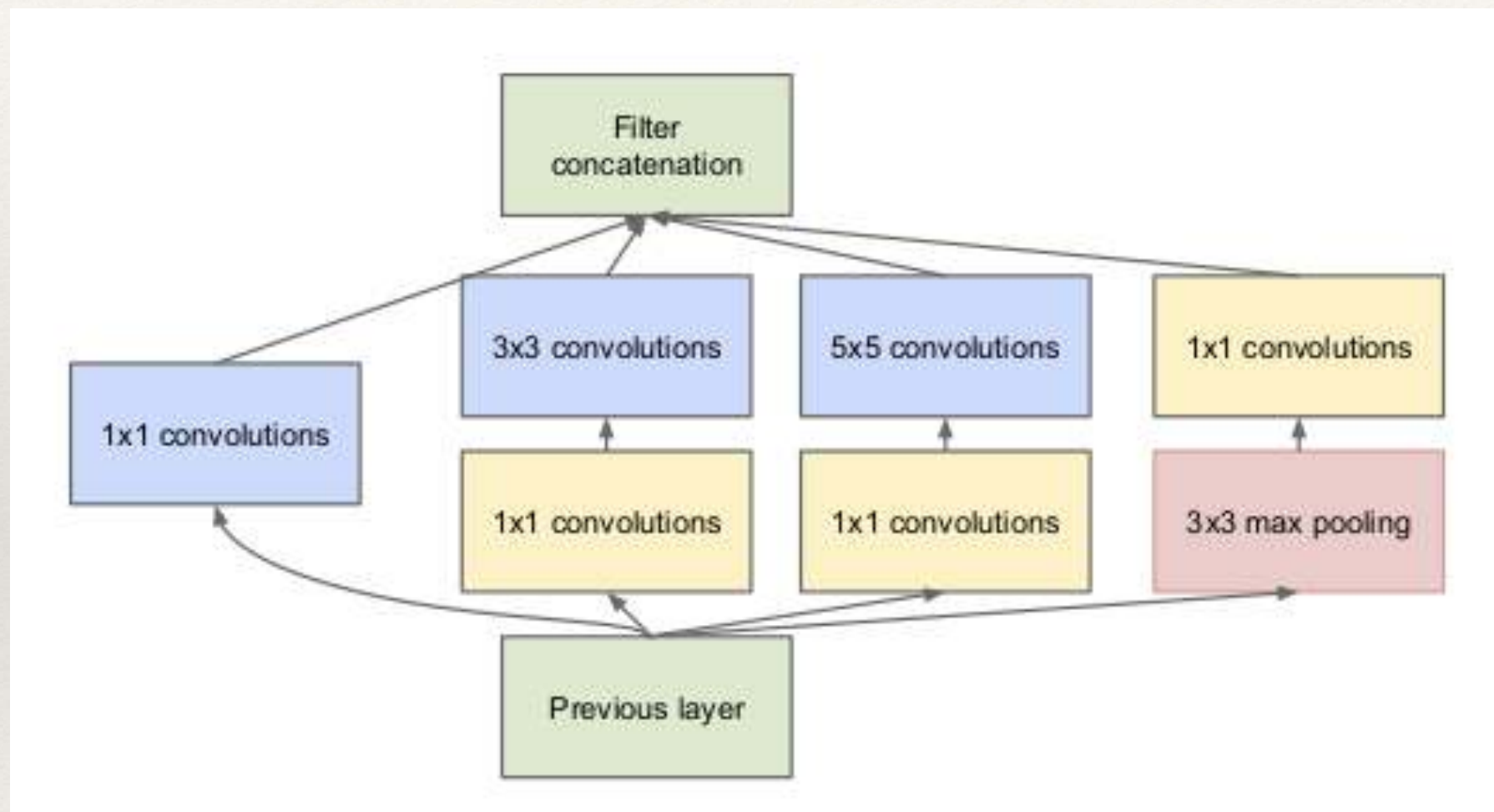


(a) Linear convolution layer



(b) Mlpconv layer

Inception Network



Mixing Convolutional and LSTM networks

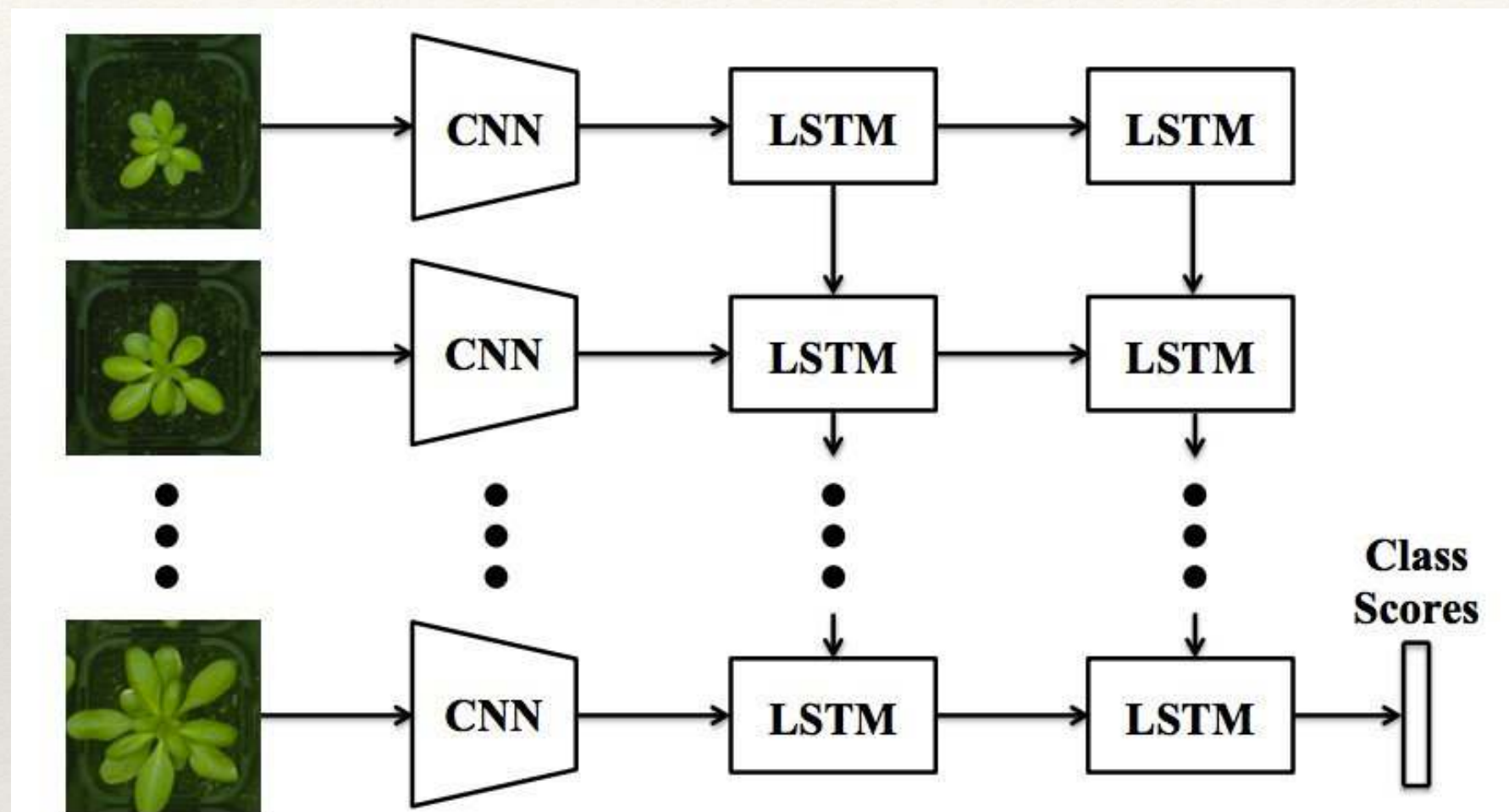


Figure 4: **The CNN-LSTM structure.** The CNNs extract deep features of the plant images and then, the growth pattern of the plant is modeled using LSTMs. Finally the genotype with highest class score is selected.

Deep Unsupervised Learning

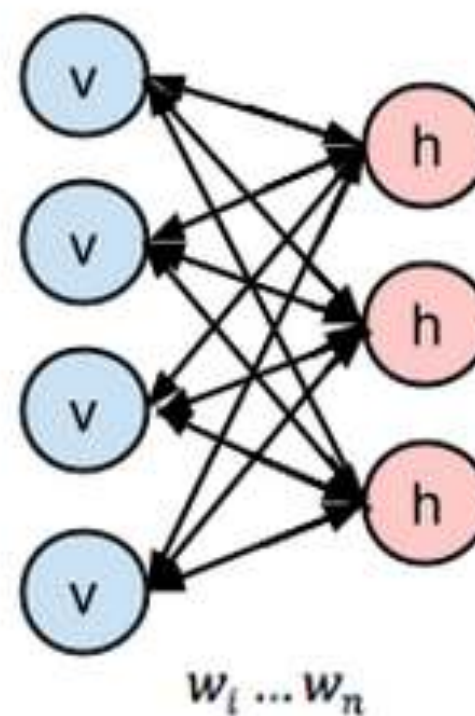
Restricted Boltzmann Machines

$$P(\mathbf{v} = \mathbf{v}, \mathbf{h} = \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{v}, \mathbf{h}))$$

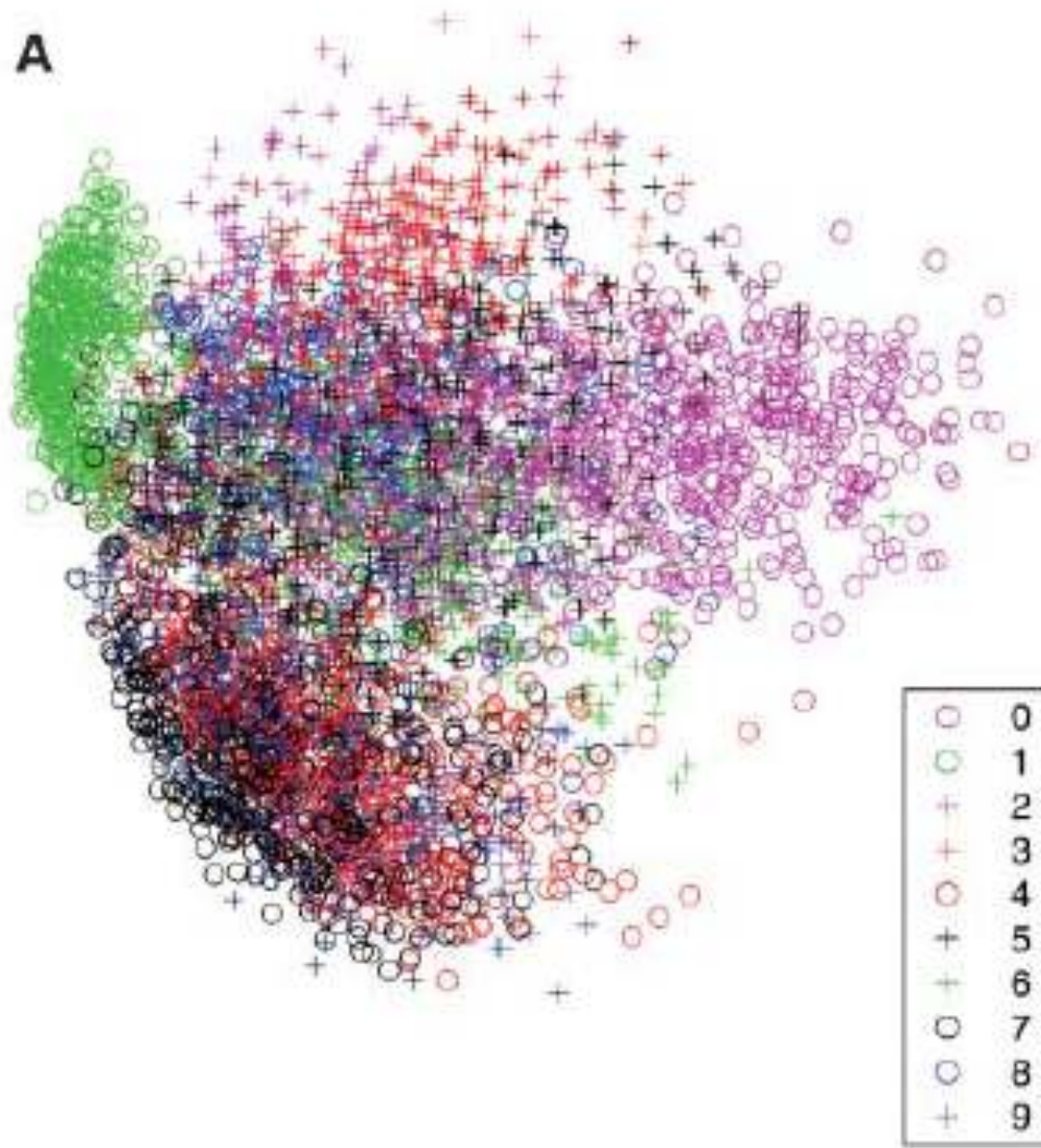
$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{b}^\top \mathbf{v} - \mathbf{c}^\top \mathbf{h} - \mathbf{v}^\top \mathbf{W} \mathbf{h}$$

$$Z = \sum_{\mathbf{v}} \sum_{\mathbf{h}} \exp\{-E(\mathbf{v}, \mathbf{h})\}$$

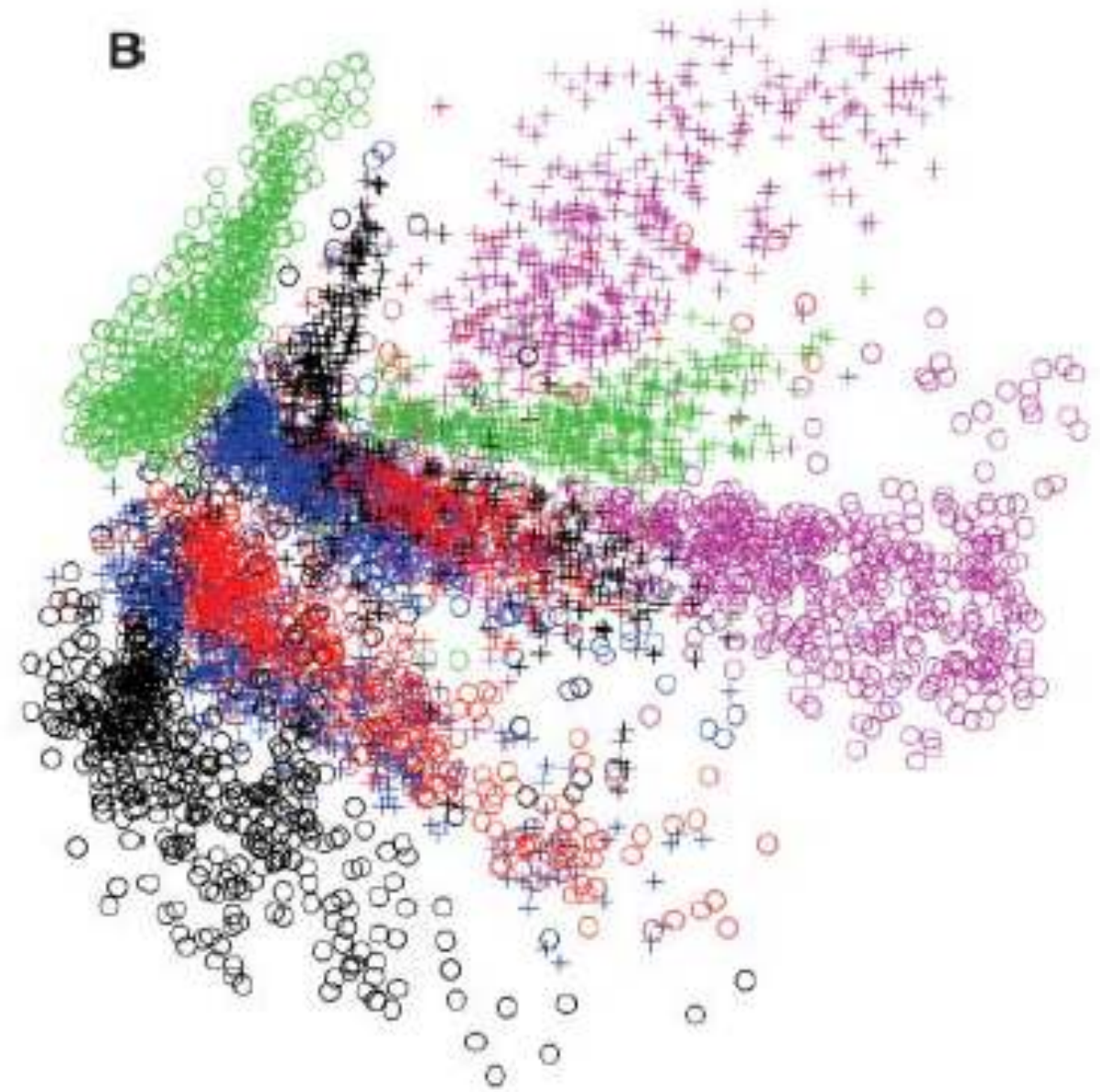
A Symmetrical, Bipartite, Bidirectional Graph with Shared Weights



Restricted Boltzmann Machine

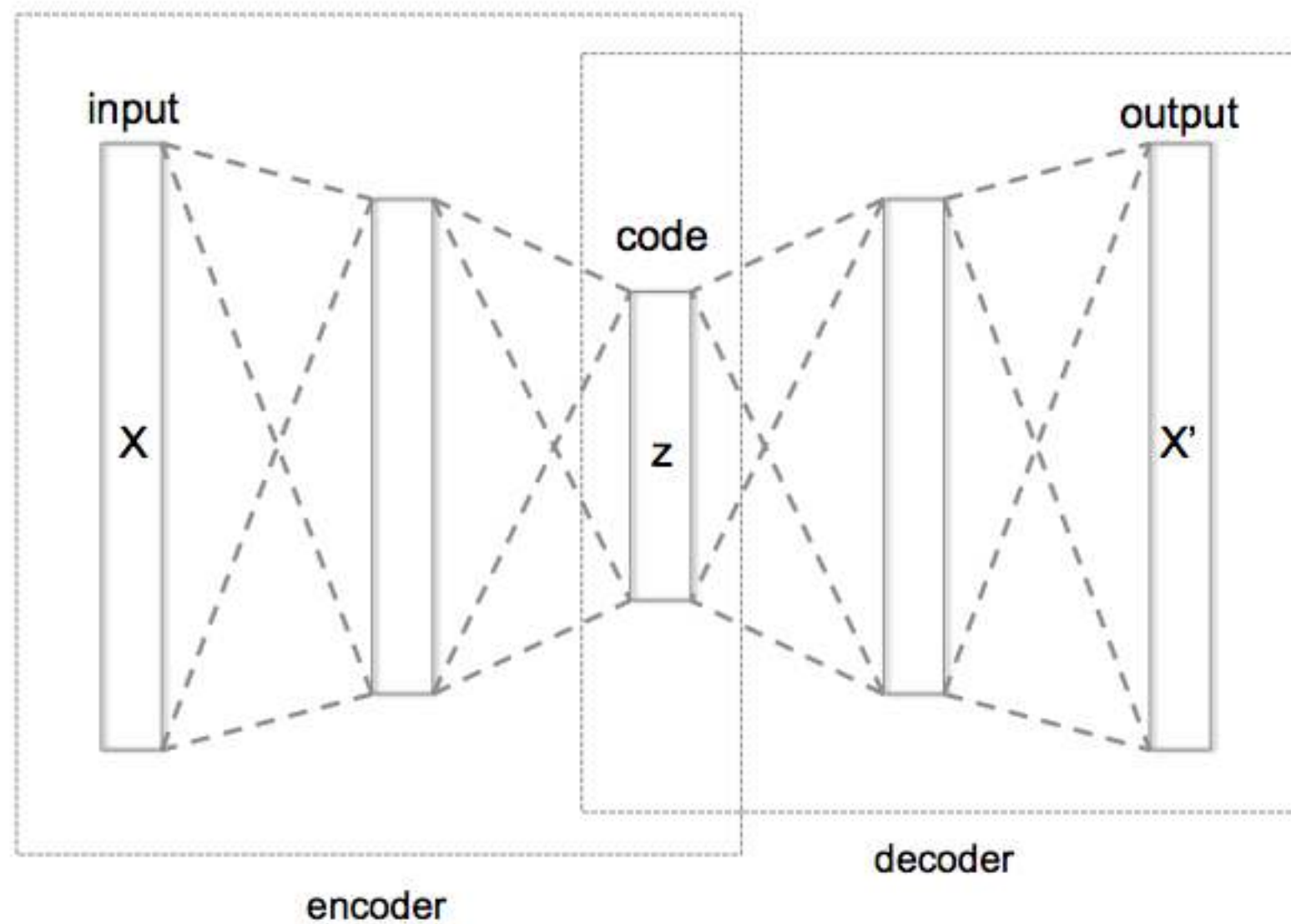


PCA



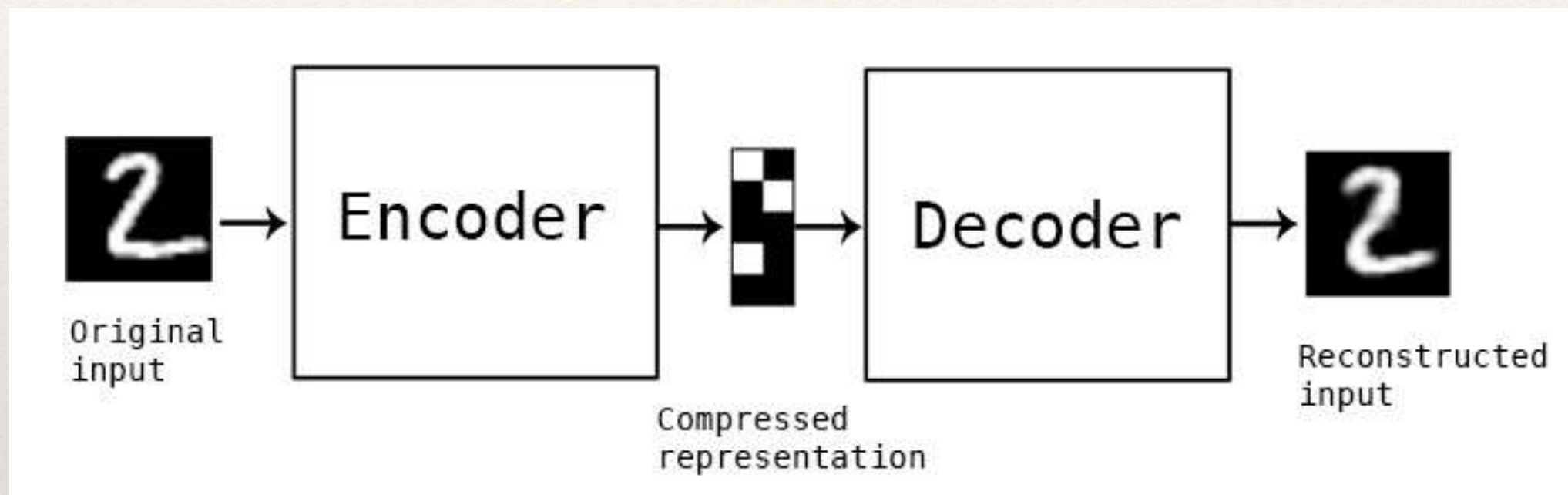
RBM

Autoencoders

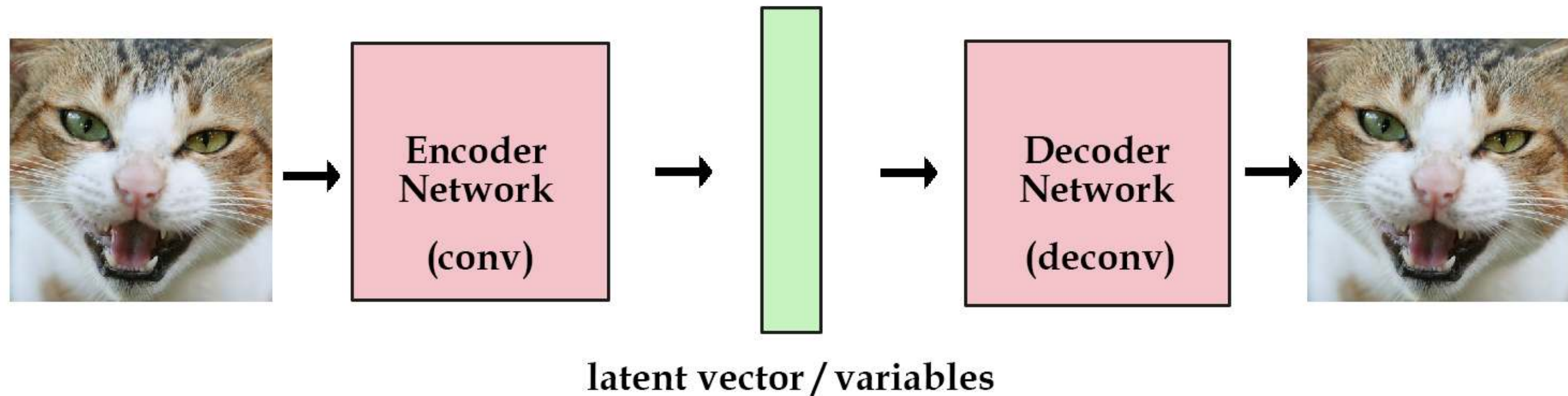


**Avoid
Identity
Mapping**

Autoencoders



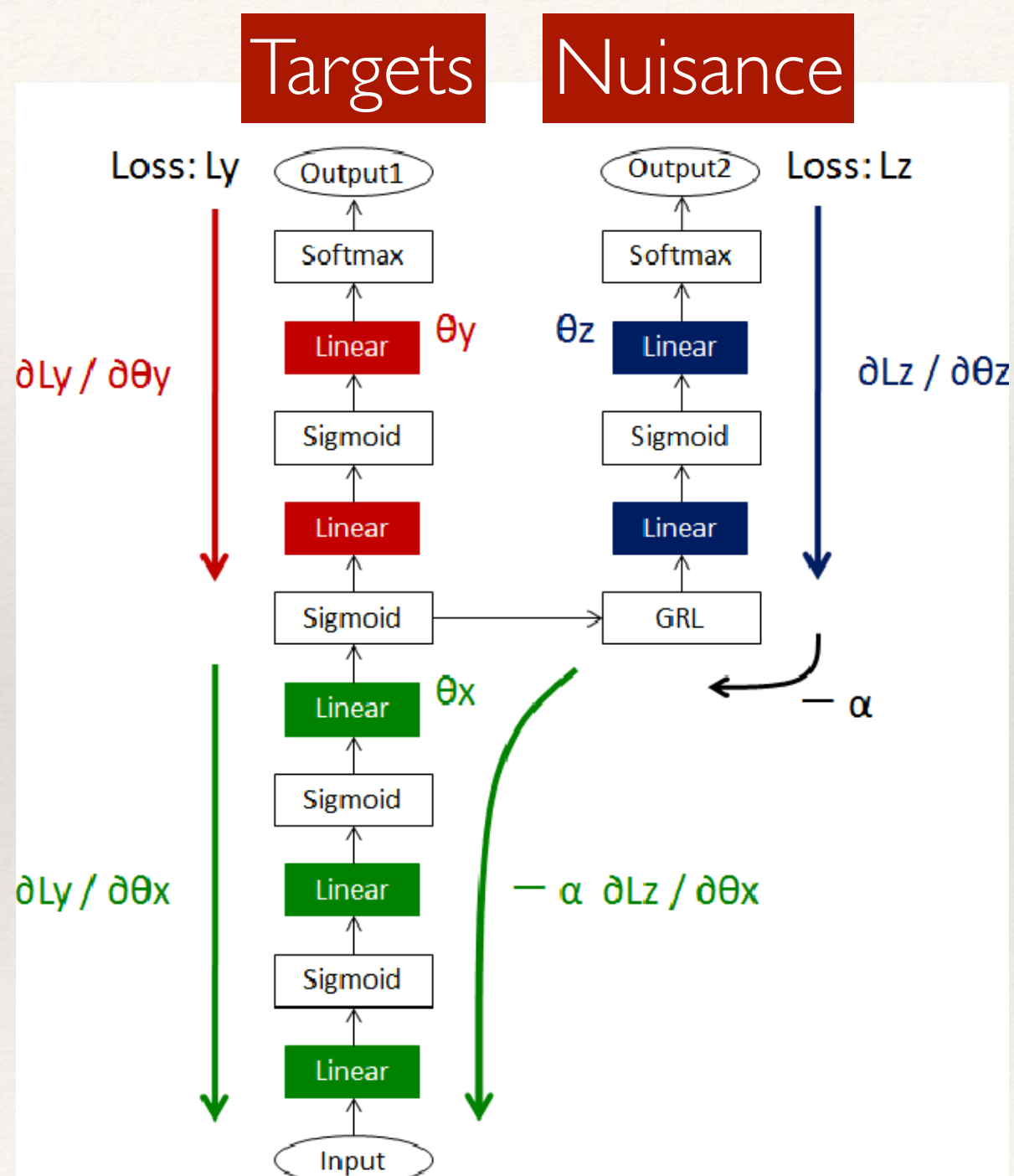
Convolutional Autoencoders



The latent vectors can form deep features for other supervised tasks.

Adversarial Learning

- The target can be learned
 - using original gradient.
 - domain adversarial gradient.
- Model will learn to be domain invariant



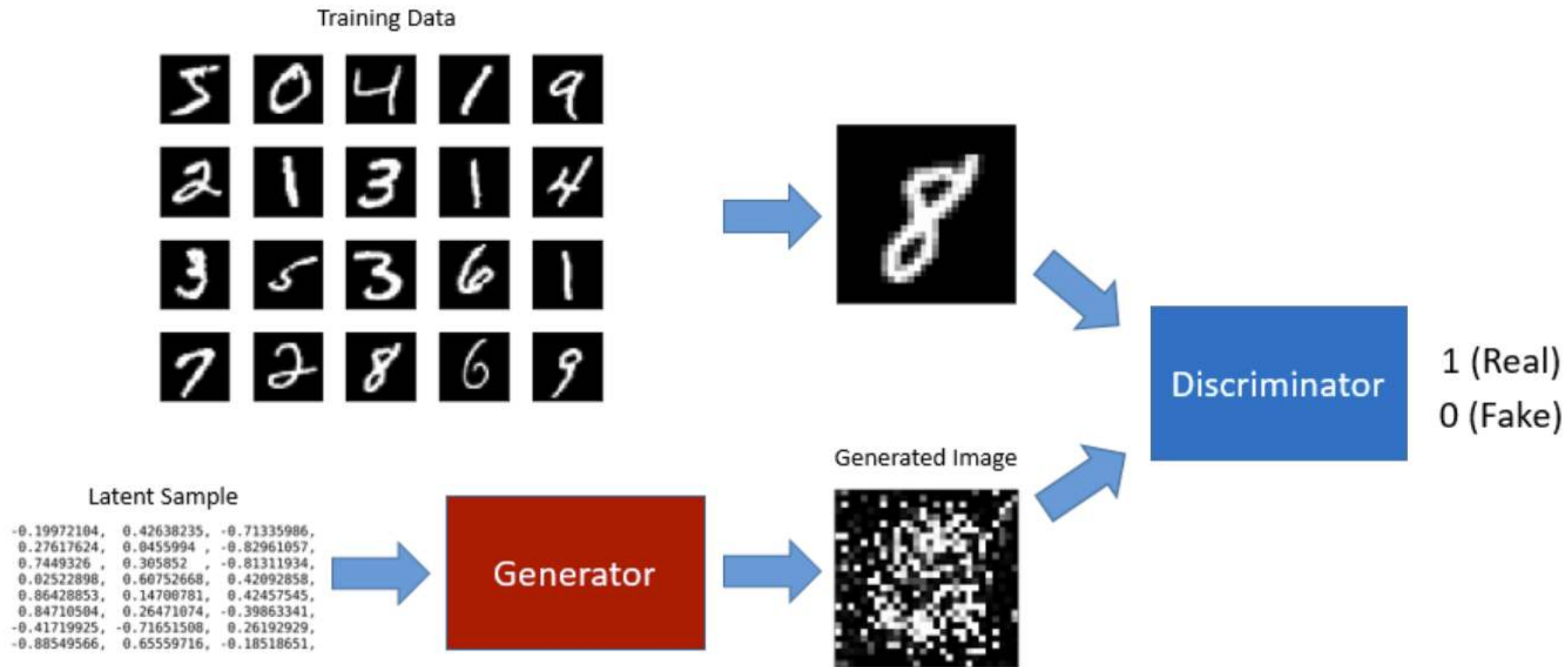
[Shinohara et al. 2017]

Generative Adversarial Networks

Generative Adversarial Networks (GANs)



Generative Adversarial Networks

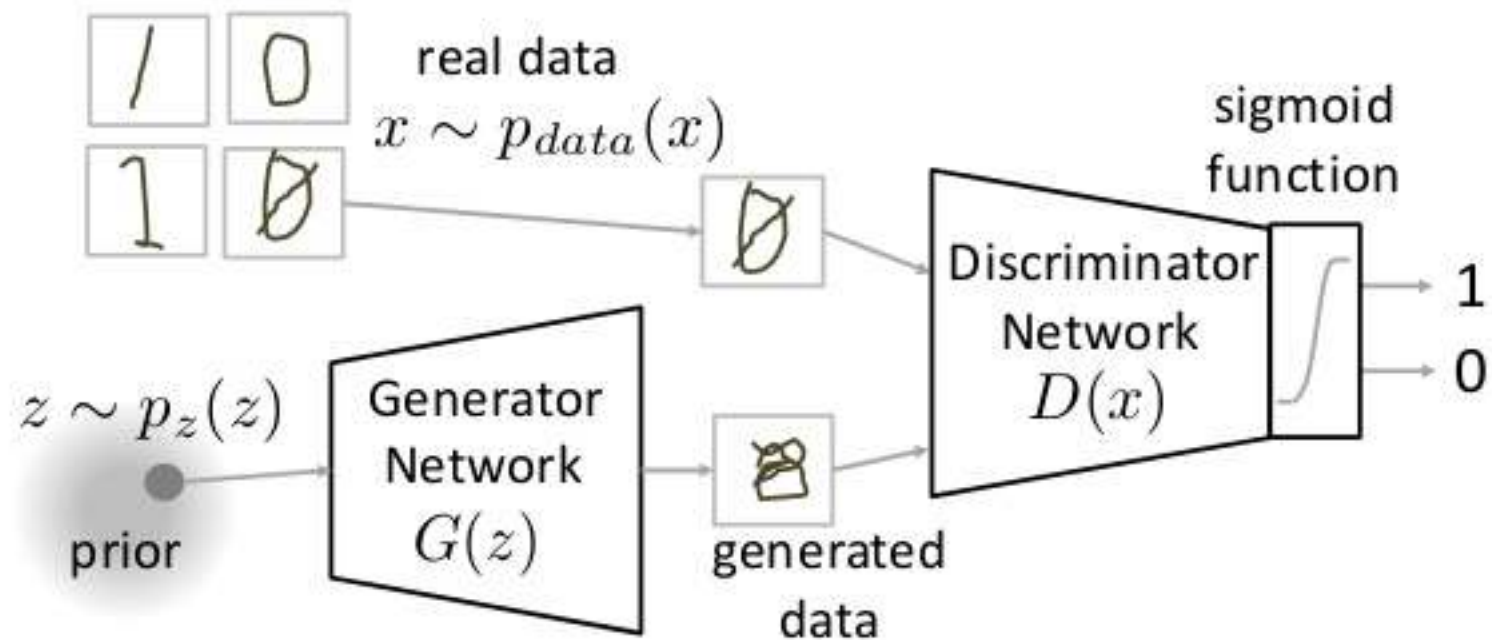


Generative Adversarial Networks

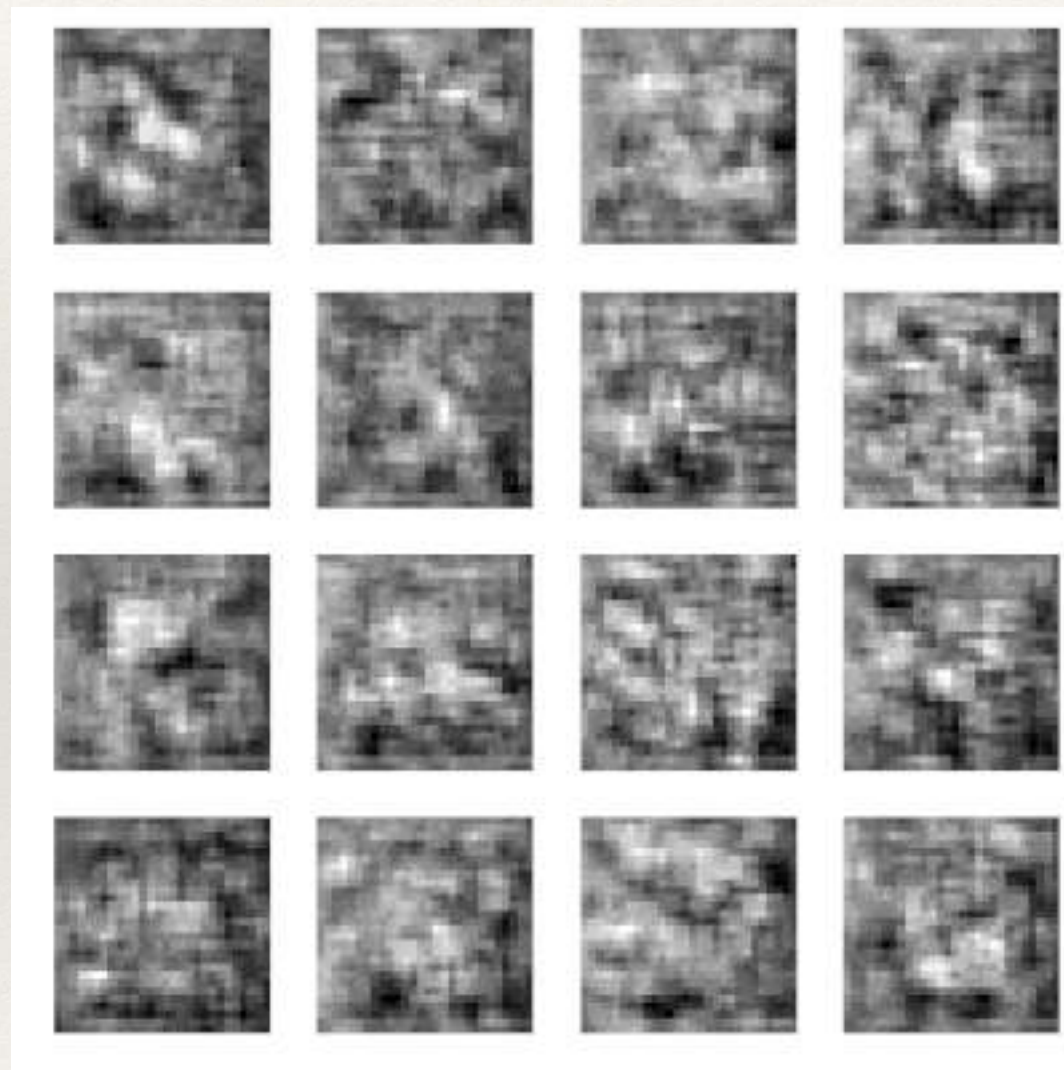
Generative Adversarial Networks

$$\min_G \max_D V(D, G)$$

$$V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

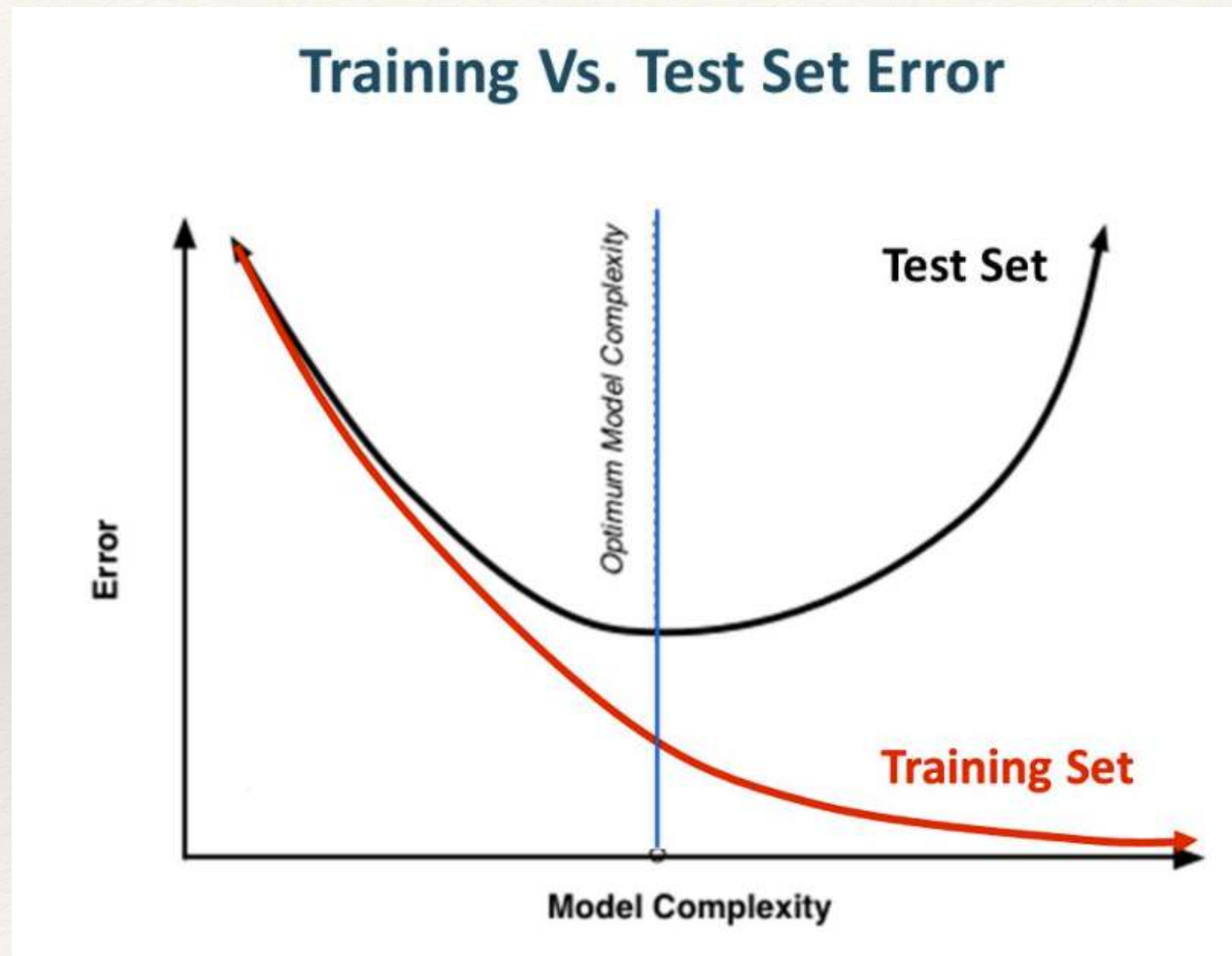


Generative Adversarial Networks

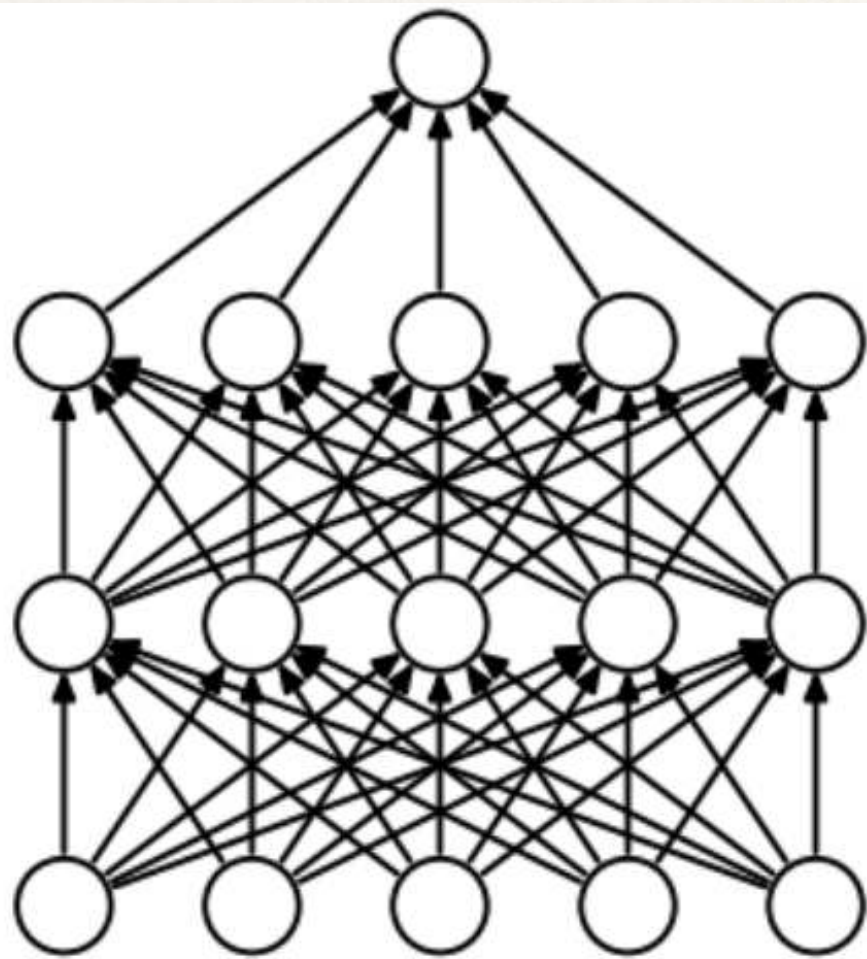


DCGANs

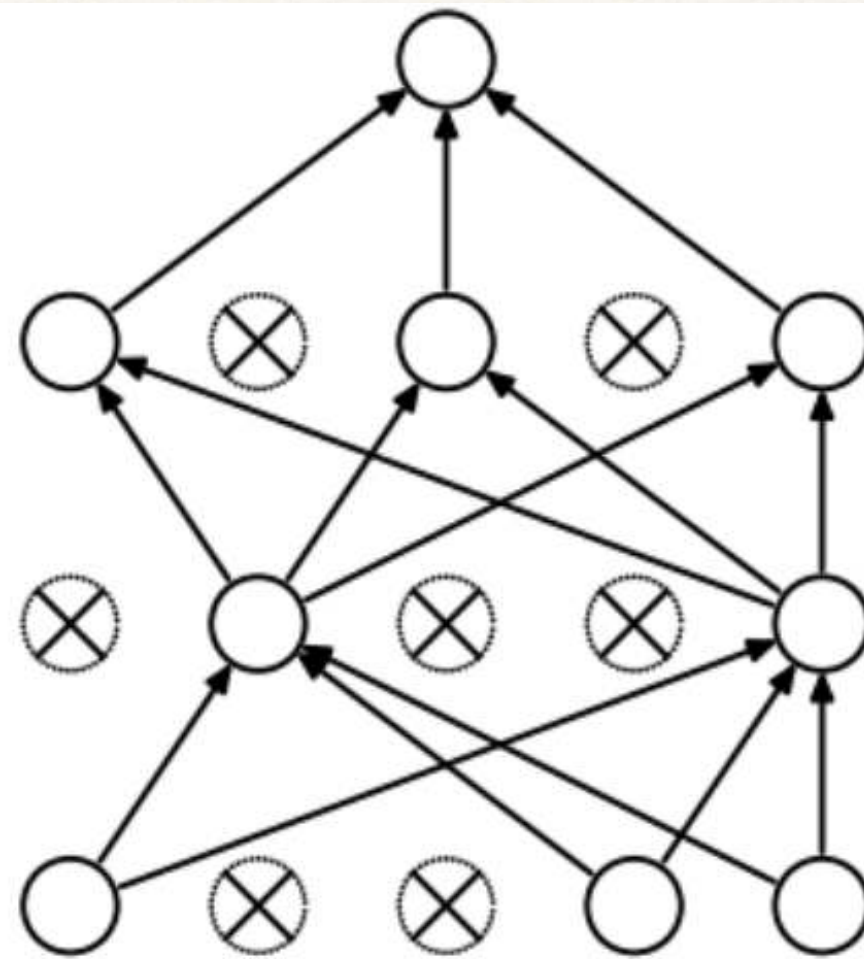
Regularization in Deep Learning



Dropout



(a) Standard Neural Net



(b) After applying dropout.